

Career Opportunities at Vietnam's First Fab

Viettel Group, Vietnam's leading technology corporation, is at the forefront of developing the national semiconductor ecosystem. Our team consists of a highly skilled workforce trained at top institutions worldwide. As part of the National Semiconductor Strategy, Viettel is committed to establishing Vietnam's first semiconductor fab by 2030.

We are seeking experienced engineers, industry experts, and outstanding young talents to join us in this pioneering endeavor. This is a unique opportunity to work in a dynamic and professional environment, where your expertise will contribute directly to shaping the future of Vietnam's semiconductor industry.

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Process Engineer

Process Engineers will directly optimize processes in Lithography, Etching, Deposition, CMP, Implantation, Cleaning, and Thermal Process. This role requires data analysis skills, process parameter control, error handling, and close collaboration with related teams to improve yield and reduce defects. Engineers will have the opportunity to develop recipes, test process improvements, and apply advanced control techniques such as APC, FDC and R2R.

Job Responsibilities

- Monitor, set up, and optimize process parameters (Recipe Development) in Lithography, Etching, Deposition, CMP, Implantation, Cleaning, and Thermal Process.
- Develop and implement advanced process control methods, such as APC, FDC, and R2R.
- Review, analyze DOE data to detect and correct errors, as well as suggest process improvements.
- Perform RCA when process errors occur.
- Participate in material and equipment parameter evaluations to ensure process compatibility.
- Develop SOP, train personnel, and support the implementation of new processes in the fab.
- Receive technology transfer within the relevant scope.

Job Qualifications

- Bachelor's or Master's degree in Semiconductor Engineering, Nanotechnology, Applied Physics, Chemistry, Electrical & Electronics, Automation, Materials Science, or related fields.
- Knowledge of one or more processes, such as Lithography, Etching, Deposition, CMP, Implantation, Cleaning, and Thermal Process.
- Strong academic background in numerical analysis, statistics, optics, chemistry, or physics.
- Proficiency in analytical tools such as JMP, Minitab, Python, and MATLAB.

Preferred Qualifications

- Ph.D. in a related field.
- 3+ years of experience in semiconductor manufacturing.
- Leadership, training, and team management skills.
- Strategic vision, the ability to provide technology consulting, and guide the company's development.

Language

- Fluent in English and Vietnamese.

Benefits

- Competitive salary determined based on the candidate's capabilities and practical experience.
- Enjoy gifts, holiday bonuses, vacation support, lunch allowance, and phone communication.
- Entitled to 12 days of annual leave, 3 days of vacation, and 1 creative day per year.
- Receive social insurance, health insurance, and unemployment insurance according to regulations, as well as additional insurance benefits from Viettel Group.
- Opportunity to work with many experts, international partners and participate in international scientific conferences in the semiconductor industry.

Equipment Engineer

Equipment Engineers play a crucial role in installing, operating, and optimizing advanced semiconductor equipment systems to ensure stability and high performance on the production lines. The role involves monitoring equipment condition, analyzing performance data, troubleshooting, and implementing advanced control systems such as Predictive Maintenance and FDC.

Job Responsibilities

- Install, monitor, maintain, and calibrate semiconductor equipment (AMHS, Lithography, Etching, Deposition, etc.), optimizing parameters to improve stability and reduce failures.
- Integrate APC, FDC systems, monitor production indicators (OEE/AUc/MQT), use SPC to detect performance issues, analyze root causes, and minimize downtime.
- Manage component inventory to ensure efficient use.
- Collaborate with suppliers to evaluate new technologies, propose upgrades, and optimize maintenance processes.
- Receive technology transfer within the relevant scope.

Job Qualifications

- Bachelor's or Master's degree in Semiconductor, Mechanical Engineering, Electrical-Electronics, Automation, Mechatronics, or related fields.
- Understanding of equipment in fabs: Lithography, Etching, Deposition, etc., and the ability to work with suppliers to address equipment-related issues.
- Experience setting up equipment parameters and implementing control systems such as Predictive Maintenance and FDC.
- Skills in analyzing equipment data, troubleshooting.
- Ability to follow detailed instructions and processes to complete tasks.

Preferred Qualifications

- Ph.D. in related fields.
- 3+ years of experience in equipment maintenance and operation in high-tech or semi manufacturing.
- Leadership skills, training, and the ability to lead an engineering team.
- Strategic vision, the ability to provide technology consulting, and guide the company's development.

Language

- Fluent in English and Vietnamese.

Benefits

- Competitive salary determined based on the candidate's capabilities and practical experience.
- Enjoy gifts, holiday bonuses, vacation support, lunch allowance, and phone communication.
- Entitled to 12 days of annual leave, 3 days of vacation, and 1 creative day per year.
- Receive social insurance, health insurance, and unemployment insurance according to regulations, as well as additional insurance benefits from Viettel Group.
- Opportunity to work with many experts, international partners and participate in international scientific conferences in the semiconductor industry.

Materials Engineer

Equipment Engineers will directly participate in controlling, optimizing the quality of materials, insulating layers, conductive metals, and surface treatment chemicals in the semiconductor manufacturing process. The role involves evaluating materials, analyzing data from SPC and DOE, implementing advanced testing methods such as XPS, XRD, TEM, SEM, EDS, collaborating with suppliers to improve quality and test new materials.

Job Responsibilities

- Inspect and evaluate input materials (wafer, photoresist, gases, and chemicals) and their impact on the process. Monitor material use in photolithography, etching, and deposition, using techniques like XPS, XRD, TEM, SEM, EDS.
- Analyze SPC and DOE data to suggest process improvements.
- Collaborate with suppliers on material quality, testing new materials, and optimizing specifications.
- Optimize material formulations and processing conditions to improve performance, reduce defects.
- Assess environmental impacts and propose solutions to minimize chemical risks.
- Receive technology transfer within the relevant scope.

Job Qualifications

- Bachelor's or Master's degree in Semiconductor, Materials Science, Chemistry, Nanoscience, Electrical-Electronics, Mechanical Engineering, or related fields.
- Understanding of semiconductor materials, insulating layers, conductive metals, and surface treatment chemicals.
- Knowledge of material analysis methods such as XPS, XRD, TEM, SEM, EDS.
- Ability to optimize material parameters and collaborate with suppliers to improve quality.
- Skills in material data analysis and product quality evaluation.

Preferred Qualifications

- Ph.D. in related fields.
- 3+ years of experience in semiconductor material research and development.
- Leadership skills, training, and the ability to lead an engineering team.
- Strategic vision, the ability to provide technology consulting, and guide the company's development.

Language

- Fluent in English and Vietnamese.

Benefits

- Competitive salary determined based on the candidate's capabilities and practical experience.
- Enjoy gifts, holiday bonuses, vacation support, lunch allowance, and phone communication.
- Entitled to 12 days of annual leave, 3 days of vacation, and 1 creative day per year.
- Receive social insurance, health insurance, and unemployment insurance according to regulations, as well as additional insurance benefits from Viettel Group.
- Opportunity to work with many experts, international partners and participate in international scientific conferences in the semiconductor industry.

Process Integration Engineer

Process Integration Engineers will directly ensure the compatibility between various semiconductor processes such as Lithography, Etching, Deposition, CMP, Implantation, Cleaning, and Thermal Process. The role includes optimizing process integration, analyzing data to improve production performance, designing and executing DOE (Design of Experiments), and collaborating with technical teams to maintain process stability and improve yield.

Job Responsibilities

- Establish and optimize processes, ensure compatibility and performance across manufacturing steps.
- Analyze large datasets to identify weaknesses in process design and issues related to equipment.
- Design, execute, and analyze DOE to evaluate the impact of individual processes on overall.
- Collaborate with technical teams to maintain process stability.
- Work with cross-functional teams to identify root causes of yield loss or defects, proposing solutions.
- Study development of advanced equipment technologies and evaluating process alternatives.
- Receive technology transfer within the relevant scope.

Job Qualifications

- Bachelor's or Master's degree in Semiconductor, Microelectronics, Materials Science, Applied Physics, Chemistry, Chemical Engineering, Nanotechnology, or related fields.
- Knowledge of semiconductor manufacturing processes (such as photolithography, etch, deposition, implant, CMP, thermal) and their interactions in integration.
- Understanding of process changes, product design/layout, and the impact of each step on yield, electrical testing, product performance, and both short-/long-term reliability.
- Experience working with FEOL, MEOL, and BEOL process flows in semiconductor manufacturing.
- Strong communication and teamwork skills.

Preferred Qualifications

- Ph.D. in related fields.
- 3+ years of experience in semiconductor manufacturing.
- Leadership skills, the ability to train and lead engineering teams.
- Strategic vision, the ability to provide technology consulting, and guide the company's development.

Language

- Fluent in English and Vietnamese.

Benefits

- Competitive salary determined based on the candidate's capabilities and practical experience.
- Enjoy gifts, holiday bonuses, vacation support, lunch allowance, and phone communication.
- Entitled to 12 days of annual leave, 3 days of vacation, and 1 creative day per year.
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Yield Engineer

Yield Engineers will directly participate in analyzing production data, identifying factors affecting yield, and implementing solutions to improve yield. The role involves statistical process control, defect analysis using advanced measurement techniques (CD-SEM, overlay, electrical testing), and close collaboration with technical teams to optimize processes and equipment.

Job Responsibilities

- Analyze production data (photolithography, etch, deposition, CMP, etc.) to identify trends, anomalies, and potential yield-impacting defects.
- Propose and implement solutions to improve wafer yield, adjusting processes, equipment, or materials.
- Use SPC to maintain production line stability, prevent errors, and collaborate with process engineers to eliminate issues.
- Install, evaluate, and maintain testing equipment, reviewing defect data.
- Work with process integration and engineering teams to identify gaps and continuous improvements.
- Knowledge of circuit design and PDK-Spice is a plus.
- Receive technology transfer within the relevant scope.

Job Qualifications

- Bachelor's or higher degree in Semiconductor, Microelectronics, Materials Science, Electrical Engineering, Statistics, or related fields.
- Deep understanding of semiconductor manufacturing processes and the factors affecting yield (including FEOL and BEOL).
- Knowledge of defect analysis and measurement techniques (CD-SEM, overlay, electrical testing).
- Experience with performance management tools (YieldStar, KLA-Tencor software and SPC tools).

Preferred Qualifications

- Ph.D. in related fields.
- 3+ years of experience in semiconductor manufacturing.
- Leadership skills, ability to train and lead engineering teams.
- Strategic vision, the ability to provide technology consulting, and guide the company's development.

Language

- Fluent in English and Vietnamese.

Benefits

- Competitive salary determined based on the candidate's capabilities and practical experience.
- Enjoy gifts, holiday bonuses, vacation support, lunch allowance, and phone communication.
- Entitled to 12 days of annual leave, 3 days of vacation, and 1 creative day per year.
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- Opportunity to work with many experts, international partners and participate in international scientific conferences in the semiconductor industry.

Device Engineer

Semiconductor Device Engineers will be directly responsible for developing and optimizing device performance. The role includes simulating and characterizing the electrical properties of semiconductor devices, analyzing data to assess reliability, and collaborating with design and manufacturing teams to enhance IC fabrication processes.

Job Responsibilities

- Analyze, simulate, and optimize the electrical characteristics of semiconductor devices (transistors, diodes, capacitors, etc.) using simulation tools such as SPICE, TCAD and Sentaurus.
- Research, develop, and optimize IC fabrication processes to enhance device performance and reliability.
- Conduct electrical measurements (IV, CV, BTI, TDDDB) to evaluate device characteristics and durability.
- Support the implementation of new technologies and verify the electrical characteristics of products.
- Work closely with design, process technology, and manufacturing teams to optimize semiconductor devices in the fab.
- Receive technology transfer within the relevant scope.

Job Qualifications

- Bachelor's or Master's degree in Semiconductor Engineering, Semiconductor Physics, Electrical & Electronics Engineering, Microelectronics Engineering, or related fields.
- Knowledge of semiconductor physics, IC fabrication, and electrical device characteristics.
- Understanding of semiconductor device characterization techniques, including IV, CV, reliability tests.
- Experience with TCAD simulation tools, such as Sentaurus and Silvaco.
- Hands-on experience with testing equipment and automated measurement systems.
- Experience in semiconductor R&D projects is an advantage.

Preferred Qualifications

- Ph.D. in a related field.
- 3+ years of experience in semiconductor manufacturing or research centers.
- Strong leadership, training, and team management skills.
- Strategic vision, the ability to provide technology consulting, and guide the company's development.

Language

- Fluent in English and Vietnamese.

Benefits

- Competitive salary determined based on the candidate's capabilities and practical experience.
- Enjoy gifts, holiday bonuses, vacation support, lunch allowance, and phone communication.
- Entitled to 12 days of annual leave, 3 days of vacation, and 1 creative day per year.
- Receive social insurance, health insurance, and unemployment insurance according to regulations, as well as additional insurance benefits from Viettel Group.
- Opportunity to work with many experts, international partners and participate in international scientific conferences in the semiconductor industry.

Quality Engineer

Quality Engineers will play a key role in building and implementing the quality control system at the semiconductor manufacturing facility. The role involves setting standards for raw materials, developing and monitoring production processes.. Engineers will apply advanced quality control methods such as SPC, DOE, and Six Sigma to reduce defects and ensure products meet international quality standards.

Job Responsibilities

- Develop quality standards for raw materials, production processes, and finished products, as well as methods to measure and evaluate product performance.
- Measure, inspect, and monitor the entire quality and reliability evaluation process, covering raw materials, wafers during production, and finished products.
- Apply advanced quality control techniques such as SPC, DOE, Six Sigma, and FMEA to manage defects and ensure product quality.
- Receive technology transfer within the relevant scope.

Job Qualifications

- Bachelor's or Master's degree in Semiconductor Engineering, Materials Science, Electronics, Chemistry, Mechanical Engineering, Nanotechnology, or related fields.
- Experience in setting quality standards and optimizing product testing processes.
- Proficiency in using DOE, FMEA, RCA, and SPC analysis tools.
- Experience with semiconductor measurement and inspection equipment.
- Familiarity with quality management systems such as ISO and ASTM standards. Certifications from reputable organizations are a plus.

Preferred Qualifications

- PhD in relevant fields.
- 3+ years of experience in quality control within the semiconductor industry.
- Leadership skills, experience training, and guiding engineering teams.
- Strategic vision, the ability to provide technology consulting, and guide the company's development.

Language

- Fluent in English and Vietnamese.

Benefits

- Competitive salary determined based on the candidate's capabilities and practical experience.
- Enjoy gifts, holiday bonuses, vacation support, lunch allowance, and phone communication.
- Entitled to 12 days of annual leave, 3 days of vacation, and 1 creative day per year.
- Receive social insurance, health insurance, and unemployment insurance according to regulations, as well as additional insurance benefits from Viettel Group.
- Opportunity to work with many experts, international partners and participate in international scientific conferences in the semiconductor industry.

Production Engineer

Production Engineers will be directly involved in establishing and optimizing operational systems at the manufacturing facility. The role includes managing production flow, optimizing factory capacity, and ensuring efficient automation systems. Engineers will apply advanced methods such as Capacity Planning, Flow Optimization, and Theory of Constraints to reduce variability and maximize throughput.

Job Responsibilities

- Understand manufacturing science concepts: Theory of Constraints, Capacity Planning, Flow Optimization.
- Ensure factory infrastructure is effectively coordinated with the production control and execution systems.
- Collect and analyze equipment data and coordination logic to optimize manufacturing operations.
- Build and maintain production scheduling systems to achieve capacity targets and reduce variability.
- Quickly address priority changes, manage multiple projects concurrently.
- Plan to maximize factory capacity to meet customer requirements.
- Manage the operations team to ensure production targets are met.
- Receive technology transfer within the relevant scope.

Job Qualifications

- Bachelor's degree in Industrial Engineering, Production Management, Systems Management Science, Mechanical & Automation Engineering, or related fields.
- Experience supervising and operating production lines, with an understanding of Throughput and Cycle Time.
- Experience implementing Capacity Planning, Material Flow Optimization, and Cost Modeling.

Preferred Qualifications

- PhD in relevant fields.
- 3+ years of experience in semiconductor manufacturing.
- Leadership skills, experience training, and guiding engineering teams.
- Strategic vision, the ability to provide technology consulting, and guide the company's development.

Language

- Fluent in English and Vietnamese.

Benefits

- Competitive salary determined based on the candidate's capabilities and practical experience.
- Enjoy gifts, holiday bonuses, vacation support, lunch allowance, and phone communication.
- Entitled to 12 days of annual leave, 3 days of vacation, and 1 creative day per year.
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- Opportunity to work with many experts, international partners and participate in international scientific conferences in the semiconductor industry.

Facility Engineer

Facility Engineers will be directly responsible for operating and ensuring that infrastructure systems meet production requirements. This includes managing design, overseeing construction, controlling quality, optimizing the operation of electrical, water, compressed air, HVAC, clean room systems, and implementing smart monitoring solutions such as SCADA and IIoT.

Job Responsibilities

- Manage the design, construction, and expansion of the factory, ensuring quality, progress, and adherence to technical standards.
- Control contractor agreements, optimize costs, and evaluate contractor capabilities.
- Supervise contractors, monitor labor safety and construction quality.
- Support testing and acceptance of construction projects.
- Assist in layout design for production equipment integration with factory systems.
- Inspect electrical, compressed air, HVAC installations, and equipment integration before production.
- Oversee infrastructure systems to ensure stable operation and minimize downtime.
- Monitor energy consumption and propose cost-saving solutions.
- Implement SCADA and IIoT to optimize infrastructure operations.
- Receive technology transfer within the relevant scope.

Job Qualifications

- Bachelor's or Master's degree in Civil Engineering, Mechanical Engineering, Project Management, Environmental Engineering, Industrial Engineering, or related fields.
- Experience in contractor management, construction project acceptance, and documentation.
- Knowledge of electrical, water, compressed air, HVAC, and clean room systems.
- Proficiency in design and monitoring tools like Microsoft Project, AutoCAD, Revit, and BIM is a plus.

Preferred Qualifications

- PhD in relevant fields.
- 3+ years of experience designing, building, and developing factory infrastructure technologies.

Language

- Fluent in English and Vietnamese.

Benefits

- Competitive salary determined based on the candidate's capabilities and practical experience.
- Enjoy gifts, holiday bonuses, vacation support, lunch allowance, and phone communication.
- Entitled to 12 days of annual leave, 3 days of vacation, and 1 creative day per year.
- Receive social insurance, health insurance, and unemployment insurance according to regulations, as well as additional insurance benefits from Viettel Group.
- Opportunity to work with many experts, international partners and participate in international scientific conferences in the semiconductor industry.

IT Engineer

IT Engineers will directly participate in building and operating the IT systems for the factory. This includes managing network systems, security, servers, MES platforms, CIM, SCADA, AMHS, FDC, optimizing data infrastructure, and developing AIoT to improve production efficiency.

Job Responsibilities

- Manage factory IT infrastructure, including networks, servers, storage, security, and automation systems.
- Ensure the operation and maintain MES, CIM, SCADA, AMHS, FDC systems, and IIoT platforms supporting production.
- Develop and optimize IT systems, including Cloud-based MES, Data Lakes, HPC, and AI/ML for big data analysis.
- Implement Predictive Maintenance, monitor equipment failures, and optimize operations based on data.
- Build a Zero Trust security architecture to protect OT/ICS systems from cybersecurity threats.
- Evaluate and implement ERP and PLM systems to optimize supply chain management and smart manufacturing.
- Develop Industrial AI and AIoT for real-time production monitoring, automation, and data analysis.
- Receive technology transfer within the relevant scope.

Job Qualifications

- Experience managing IT infrastructure, including networks (LAN, WAN), servers, storage, and system security.
- Understanding of MES, CIM, SCADA, AMHS, FDC, and factory automation systems.
- Experience with Data Lake, HPC and AI/ML in production data analysis.
- Knowledge of cybersecurity, OT/ICS protection, and Zero Trust architecture implementation.

Preferred Qualifications

- PhD in related fields.
- 3+ years of experience managing modern factories or semiconductor factories.

Language

- Fluent in English and Vietnamese.

Benefits

- Competitive salary determined based on the candidate's capabilities and practical experience.
- Enjoy gifts, holiday bonuses, vacation support, lunch allowance, and phone communication.
- Entitled to 12 days of annual leave, 3 days of vacation, and 1 creative day per year.
- Receive social insurance, health insurance, and unemployment insurance according to regulations, as well as additional insurance benefits from Viettel Group.
- Opportunity to work with many experts, international partners and participate in international scientific conferences in the semiconductor industry.